

Suho Yu

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Research Interest

UAM • eVTOL system integration • High-voltage battery architecture • Autonomous aerial robotics
Embedded control • Fault-tolerant design • Certification & safety • HIL testing • Edge AI using Jetson • MDO

Education

Inha University

BS in Mechatronics Engineering

Mar 2018~Aug 2023

Incheon, KR

Courses Control Systems, Mechatronics Design, UAV System Engineering
Embedded Systems, Signal Processing, Dynamic Systems

Research Experience

eVTOL Integration	Intelligent On-board Avionics SW for Emergency Detection & Response <i>VSPACE, System Lead – Fault detection avionics software integration</i>	Apr 2024~Present
	Urban Air Mobility (UAM) Type Certification Research <i>VSPACE, Propulsion Integration Lead - Propulsion architecture design</i>	Apr 2024~Present
	Future Air Mobility UAM Prototype – Development and Flight <i>VSPACE, Project Lead – System architecture & experimental validation</i>	Jan 2022~Dec 2024
Battery System & Integration	Korea's First Manned All-Electric GA Aircraft Flight – Full Technical Research <i>VSPACE, Battery System Lead – High-voltage pack design and integration</i>	Apr 2020~Dec 2023
	Development of UAV System for Maritime Safety and Ecosystem Monitoring <i>VSPACE, Battery System Lead – Electric propulsion system integration</i>	Apr 2021~Sep 2024
	Development of 250 Wh/kg-class battery system for UAM applications <i>VSPACE, Principal Investigator – High-voltage pack design and integration</i>	Apr 2020~Dec 2023
	High-discharge Battery System for 200 kg Hydrogen-Electric Cargo UAV <i>VSPACE, Battery System Lead – High-voltage pack design and integration</i>	Apr 2021~Sep 2023
	Development of Battery System for 40kg-class Heavy-Lift Multirotor Drone <i>VSPACE, Battery System Lead – High-voltage pack design and integration</i>	Apr 2021~Sep 2023

Publication

eVTOL Integration	Design of a Power System for Handling Regenerative Braking Back EMF in a 1000 kW Class Electric Test Facility <i>SASE 2025 Fall Conference – First Author</i>	Sep 2025
	Integrated System Design and Flight Demonstration for Ensuring Flight Continuity in an X8 Electric Propulsion Aircraft under Single Propeller Failure <i>Journal of Aerospace System Engineering – First Author</i>	Apr 2025
	A Methodology for Specifying Electric Propulsion Battery Requirements Based on K-UAM Operational Scenarios <i>Journal of Aerospace System Engineering – Co-Author</i>	Apr 2025
	Development of Manned Electric Propulsion LSA Aircraft Modification <i>Journal of Aerospace System Engineering – Co-Author</i>	Jun 2023
Battery System & Integration	Development of 230 wh/kg High-Power, High-Density Battery System <i>Journal of Aerospace System Engineering – Co-Author</i>	May 2023
	Effect of Internal Resistance Deviation on Thermal Performance <i>Korean Society for Automobile Safety – First Author</i>	May 2023
	Thermal Analysis of the high C-rate Battery Module in an Urban Air Mobility <i>The Korean Society of Automotive Engineers – Co-Author</i>	Nov 2022

Patents

eVTOL Integration	Device and Method for Detecting Anomalies Using Multivariate Sensor Data by Flight Phase of a Drone 10-2025-01527715-KR	Nov 2025
	Vibration-Absorbing Mounting Structure for Flight Control Computer 10-2025-01527748-KR	Nov 2025
	Device and Method for Detecting Anomalies Using Multivariate Sensor Data by Flight Phase of a Drone 10-2025-01527715-KR	Nov 2025
	Apparatus and Method for Active Output Control Responding to Abnormal State of a Flying Object Power System 10-2025-01527715-KR	Nov 2025
	Method and Apparatus for Controlling Flight Vehicle Landing Based on Multi-Sensor Reliability 10-2025-01527748-KR	Nov 2025
	Flight Control Computer Housing With Integrated Heat Dissipation and Electromagnetic Shielding Functions 10-2025-01527715-KR	Nov 2025
	Flight Control Computer Having Redundant CAN Communication Structure and Operating Method Thereof 10-2025-01527748-KR	Nov 2025
	Method and Apparatus for Automatic Power Source Switchover With Multiple Inputs and Flight Control Computer Including the Same 10-2025-01527748-KR	Nov 2025
	Urban Air Mobility Frame 15366-000102-US	Jul 2025
Battery System & Integration	Battery Module for Electric Propulsion Aircraft and Battery Pack Including the Same 10-2025-01527733-KR, U.S. Patent Pending	Oct 2025
	High-Power Battery Package for eVTOL and Its Management Methods 10-2024-0200697-KR	Dec 2024
	Air-Cooled High-Density Battery Pack 10-2024-017151-KR	Dec 2024
	Aircraft Battery Management for Takeoff and Landing 10-2024-004711-KR	Jan 2024
	Battery Management According to Flight Plan 10-2024-0047710-KR	Jan 2024
	Battery Management Method for Urban Air Mobility 10-2024-0047710-KR	Jan 2024
	Movable Charger 10-2023-0163497-KR	Nov 2023
	Fire Protection Pad and Battery Cell Array using the Same 10-2023-0054985-KR	Apr 2023
	Two-Wheeled Vehicle Vibration Dampers KP 21047	May 2021

Professional Experience

VSPACE Co., Ltd. <i>Director of Engineering</i>	Nov 2021~Present Incheon, KR
CEAS Co, Ltd. <i>Design Engineer</i>	Jul 2018~Jan 2021 Ansan, KR
ROBO Co., Ltd. <i>Production Manager</i>	Mar 2017~Jul 2018 Seoul, KR
SEC Co., Ltd. <i>Design Engineer</i>	Jan 2016~Feb 2017 Suwon, KR

Awards & Achievement

Aerospace Technology-Based Start-Up Academy, 2022 – Best Award <i>Korea Aerospace Research Institute</i>	Oct 2022
CNC Milling, 2015 Korea National Skills Competition – Gold Medal <i>Korea Skills Promotion Committee</i>	Apr 2015
CNC Milling, 2023 Korea National Skills Competition – Judge <i>Korea Skills Promotion Committee</i>	Oct 2023
CNC Milling – Player Coaching (192hr) <i>Korea Aviation High School</i>	Dec 2023~Feb 2024

Skills

Tools:	SolidWorks, CATIA V5, AutoCAD, Master CAM, STAR-CCM+, Altium Designer STM32CubeIDE
Programming:	C, C++, Embedded C, Python
Certification:	Engineer – General Machinery, Industrial Safety, Electricity, 3D Printer Development, Computer Aided Manufacturing Craftsman – Computer Aided Milling, Computer Aided Lathe, Fork Lift Operator, Hydro-pneumatic, 3D Printer Operation
Languages:	Korean (Native), English (Advanced), Japanese (Advanced)